

Delphes vs. CMS for $HH \rightarrow b\bar{b}\tau\tau$:

where we stand, and one decision to take

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Abstract

The $\tau_h\tau_h$ yield deficit that motivated this work (a factor ~ 5) is closed: the cutflow agrees with CMS at every stage (1.01) and eight of the ten NSBI features agree within statistics. Two do not — $m_{\tau\tau}$ and $\Delta R_{\tau\tau}$ — and both are on the τ side. The $m_{\tau\tau}$ offset has now been traced: it is in the τ four-vectors rather than the mass estimator or the p_T^{miss} , fake τ_h are excluded as its cause by measurement, and all of it sits in the *gen-matched* population. Since the energy-scale map is a per- p_T *median* ratio, it can align the peak but cannot reshape a response distribution — so the residual may lie outside what the correction approach can reach at all. That would turn the open decision (Section 3) from a question of elegance into one of feasibility. One measurement settles it, and it has not been run yet.

1 Status

The card is the stock CMS card plus the patches in `docs/card_patch_validation.md`. All object tuning is applied *downstream*, so the produced samples are never re-made; every efficiency and scale is measured on a private 2024 NanoAODv15 anchor, so “tuned” below always means “measured on CMS and transferred”.

quantity	status	how
b-tag efficiency (b , c , light)	✓ tuned	stochastic re-tag from <code>Jet.Flavor</code> at the anchor-measured UParT-AK4 Medium efficiency
b-jet energy scale	✓ tuned	response inverted to unity against <code>GenJet</code> (self-anchored, not like-for-like)
m_{bb} peak and width	✓ matched	follows from the above
τ_h identification efficiency	✓ tuned	anchor <code>GenVisTau</code> → <code>DeepTau</code> v2.5 Medium; includes CMS’s HPS reconstruction inefficiency
τ_h visible mass	✓ tuned	<i>sampled</i> per jet from the anchor’s per- p_T mass quantiles
τ_h acceptance	✓ identical	common $p_T > 20$ GeV, $ \eta < 2.3$
jet- τ overlap removal	✓ identical	di- τ pair chosen first, then $\Delta R > 0.4$
p_T^{miss} resolution	✓ tuned	Gaussian smearing to the anchor resolution ($16.5 \rightarrow 32.6$ GeV per component)
lepton reconstruction efficiency	✓ tuned	per- p_T scale factors ($1.57 \rightarrow 1.22$ for e , $1.30 \rightarrow 1.13$ for μ)
di- τ mass estimator	✓ identical	the same covariance-free FastMTT on both sides
τ_h energy scale	(✗) tuned	CMS/Delphes response ratio against a <i>visible</i> τ reference — but it under-corrects; Section 2.1

Table 1: Object-level status. Every map is derived on the anchor and applied downstream; none requires a change to the card.

feature	status	comment
m_{HH}	✓	the κ_λ -carrying observable; agrees across $\kappa_\lambda = 0, 1, 5$
m_{bb}	✓	both peak near 112 GeV
$p_T^{HH}, \Delta\phi_{HH}, p_T^{H1}, p_T^{H2}$	✓	
ΔR_{bb}	✓	after symmetric overlap removal on both sides
$\cos\theta^*$	✓	statistically limited but consistent
$m_{\tau\tau}$	✗	~ 7 GeV high, Delphes broader; Section 2.1
$\Delta R_{\tau\tau}$	✗	Delphes excess at 2.2–3.2; Section 2.2

Table 2: NSBI inputs at $\kappa_\lambda = 0$; the other κ_λ points behave the same. Yields: gen-matched cutflow 1.01, analysis-level selection $1.18\times$.

Known limitations, accepted for now. None of these is implicated in the two disagreements above, but they bound what the samples can be used for.

- **The τ_h fake rate is measured on signal.** `tau_mistag` comes from the b -rich HH anchor. Fake rates are strongly jet-flavour dependent, so this value is *not* appropriate for the fake-dominated backgrounds (QCD, W+jets), which need their own measurement before they are trusted.
- **p_T^{miss} smearing is isotropic and flat**, where the real resolution is anisotropic and grows with ΣE_T . This reproduces the width but not its structure.
- **Pileup enters only through p_T^{miss} .** The card runs without pileup, so jet energy *resolution* is untuned and pileup jets cannot be produced at all. Both act on the $b\bar{b}$ side, which agrees; the second cannot be fixed downstream in any case.
- **Two defects in the energy-scale application**, real but not the cause of the $m_{\tau\tau}$ offset: fake τ_h match neither the b nor the τ branch of `escale_factor` and are left uncorrected while still receiving a τ -like mass; and the τ branch matches *all* gen τ (`|pid|==15`) including leptonic ones, whereas the map is derived on the hadronic visible set.
- **The sampled τ_h mass is drawn independently** of the τ energy and of the partner leg, whereas the true visible mass correlates with the decay mode and hence with prong multiplicity and energy sharing. Unmeasured; distinct from the quantile-grid defect below, which *was* measured.
- **FastMTT is covariance-free** (a fixed p_T^{miss} resolution rather than the per-event covariance). Run identically on both sides, so it largely cancels in the comparison; it limits the absolute $m_{\tau\tau}$ resolution, not the agreement.
- **Every map is flat-extrapolated above 300 GeV**, the last bin edge. Untested, and worth checking: κ_λ discrimination lives at high m_{HH} and therefore at high jet p_T .

2 The two open questions

2.1 $m_{\tau\tau}$: the offset is in the τ four-vectors

Because $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ factorises, the *visible* di- τ mass m_{vis} — built from the two selected τ four-vectors, with no p_T^{miss} and no FastMTT — separates a τ energy-scale error from an error in the x fit. The two have disjoint fixes, so it is worth measuring before tuning anything.

What is measured. On the $\kappa_\lambda = 0$ overlay, m_{vis} is offset by *at least* as much as $m_{\tau\tau}$ is (peak ratios ≈ 1.24 against ≈ 1.17 , read off the histograms). The τ four-vectors therefore carry the discrepancy and the FastMTT fit is not its source; if anything it compresses it. Both τ legs are

individually harder on Delphes. Consistently, with an applied scale of $\simeq 0.85$ and a residual ratio of $\simeq 1.24$, the raw Delphes τ are $\sim 46\%$ harder than CMS while the map corrects only 15%.

Splitting into gen-matched and fake components then **excludes fakes**: the fake fractions are 4.1% on Delphes against 4.3% on CMS with indistinguishable shapes, while the gen-matched component carries the whole discrepancy — both the peak displacement and an unphysical tail at $m_{\text{vis}} > 125 \text{ GeV}$, which is kinematically impossible for $H \rightarrow \tau\tau$ and which CMS does not populate. The same pattern holds for $\Delta R_{\tau\tau}$ (Section 2.2); both τ -side disagreements live in *real* τ .

Why this may be a statement about the representation. `tau_scale` is a per- p_T -bin *median* ratio applied multiplicatively. By construction it aligns medians; it cannot change the *shape* of the response distribution. An $R = 0.4$ cone absorbs underlying event and nearby particles for some τ , where an HPS signal cone would not, producing a one-sided high tail in the response. Such a tail is not reachable by any multiplicative median map — one can move the peak into place and still be left with entries in a kinematically forbidden region, because a scale factor cannot depopulate one. If that is what is happening, it is a hard floor on the correction approach rather than a defect in its implementation.

The measurement that settles it (not yet run). Plot the τ energy response — reco τ p_T over gen-visible- τ p_T — as a *distribution* on both sides rather than as a per-bin median. A longer Delphes high tail at matched median confirms the paragraph above and makes the case for rebuilding. Agreeing shapes would instead indict the derivation, with selection bias the prime suspect, since the response is measured only on τ that were both reconstructed *and* passed ID.

What has been excluded. (i) *The sampled τ_h mass.* The FastMTT floor $x_{\min} = (m_{\text{vis}}^{\text{leg}}/m_\tau)^2$ exceeds a typical fitted $x \simeq 0.5\text{--}0.9$ only once the leg mass passes $\sim 1.26 \text{ GeV}$, so below that the assigned mass is inert: displacing the entire low-mass region by 200 MeV moves $\langle m_{\tau\tau} \rangle$ by 0.00%, against -2.4% for the same displacement in the a_1 tail. Replacing the 21-quantile grid with an explicit atom at $m_{\pi^\pm}$ plus a histogram CDF — more faithful to a distribution that genuinely has an atom — changes $\langle m_{\tau\tau} \rangle$ by 0.11%. (ii) *A symmetric resolution mismatch.* Widening the response from 7% to 25% at fixed median moves the mean by 0.05% (though it broadens the $m_{\tau\tau}$ RMS by $\sim 2\%$). Note the scope: that test used a roughly symmetric log-normal widening, so it does *not* cover the one-sided tail above. Of everything tested, only the energy *scale* — linear in $m_{\tau\tau}$ — and a one-sided response tail remain capable of what is observed.

2.2 $\Delta R_{\tau\tau}$: a selection effect, cause unknown

Comparing the *generator-level* $\Delta R_{\tau\tau}$ and the selection efficiency against it: the gen spectra agree, so the samples are the same physics and no generator-level explanation is available. The efficiencies differ — Delphes selects 0.22–0.23 of events at gen $\Delta R > 2.5$ where CMS selects 0.15–0.17 ($\sim 3\sigma$), with a weaker ($\sim 2\sigma$) hint that Delphes also loses pairs below $\Delta R < 1$. Both push the normalised distribution rightward. Three explanations were proposed and excluded by measurement: fake τ_h (the discrepancy lives entirely in the gen-matched component); τ -pair selection (making it fully symmetric moved $\Delta R_{\tau\tau}$ but not $m_{\tau\tau}$); and a large CMS τ energy miscalibration (the anchor response is 0.97–0.99).

This one matters more than its size suggests: it is a *selection* effect, not a scale, so no downstream correction can reach it. Unlike $m_{\tau\tau}$, we have no candidate mechanism.

3 The decision to think about: keep correcting the jet, or rebuild the τ_h ?

A τ_h in CMS is an HPS object built from a narrow set of decay products; in Delphes it is an $R = 0.4$ jet carrying `TauTag`. Its mass, energy and direction are therefore all wrong at source, and

we correct the first two against the anchor rather than fixing the representation. The alternative is to rebuild it from the Delphes particle-flow candidates, which the card already writes (the three EFlow collections — tracks, photons, neutral hadrons — kept, per the card header, precisely for “ τ redefinition”): seed from a jet, collect charged hadrons and photons in a narrow signal cone (fixed $\Delta R \simeq 0.1$, or p_T -dependent $3.0/p_T$ bounded at 0.05–0.1), form the standard decay-mode hypotheses, take the four-vector as the sum of signal-cone constituents, and use an isolation annulus as a fake handle.

In favour. Mass, energy and direction become correct *by construction* rather than by correction. For a phenomenological paper that removes a fast-simulation limitation instead of documenting one. On cost, the implementation is contained and is *not* on the critical path — running the full 5×10^5 -event samples through the ladder and training the NSBI is the multi-day item.

The decisive argument, *if* the response measurement of Section 2.1 confirms it, is stronger than elegance: a multiplicative per- p_T median cannot depopulate a kinematically forbidden region, so no refinement of the current maps would close the gap. That is the one argument here a referee could not answer with “tune it better”.

What does *not* argue for it. Rebuilding is sometimes said to retire three corrections at once — the visible-mass map, the energy scale and the visible- τ reference. That is badly weighted. The visible-mass map’s residual imperfection is worth 0.11% on $\langle m_{\tau\tau} \rangle$, so retiring it buys almost nothing: the map already does the work. The visible- τ reference is a derivation-side simplification, not a per-event gain. Effectively one of the three matters, and it is the energy scale.

Against, or at least uncertain. It cannot reproduce DeepTau, so the identification and fake-rate maps remain (they already work). It is *not* obviously the cure for $\Delta R_{\tau\tau}$: CMS HPS also seeds from AK4 jets, so the merging scales are similar and a narrow signal cone may not change which pairs are resolved. And the current treatment already agrees on eight of ten features and on the yield.

The caveat we most want checked. Delphes neutral particle-flow candidates are *calorimeter towers* ($\sim 0.02 \times 0.02$ in the barrel), not clustered PF objects. Since π^0 content sets both the visible mass and the decay-mode assignment, an HPS-like reconstruction on Delphes would be approximate in exactly the place that matters most — possibly no better than the anchor-sampled mass already in use.

Questions for τ expertise.

1. Is tower-granular neutral reconstruction adequate for a useful decay-mode assignment, or does it degrade the π^0 content enough that the rebuilt mass is no better than the anchor-sampled one?
2. Signal cone: fixed 0.1 or p_T -dependent, over the 20–60 GeV range that dominates here?
3. Seeding: jets (as CMS HPS does) or direct clustering of PF candidates? Would either change the efficiency for collimated pairs, $\Delta R_{\tau\tau} \lesssim 1$?
4. Is the $\Delta R_{\tau\tau}$ pattern — over-efficient above 2.5, possibly under-efficient below 1 — recognisable as a jet-versus-HPS effect at all, or should we look elsewhere?
5. Is a jet-based τ_h with anchor-measured corrections defensible in a phenomenological paper, or is rebuilding the expected standard?
6. Is a 1.18 $\times$ analysis-level yield excess acceptable for an unbinned NSBI study, where training uses shapes and the normalisation is fitted?

A Reference: the studies behind these numbers

All reproducible from the `delphes-pipeline` repository.

1. **$\tau_h\tau_h$ cutflow** (`tau_h_cutflow.py`). Five stages — gen visible τ_h , reco object, ID, cleaned jets, FastMTT — walked identically on both samples, comparing *conditional* efficiencies. Initially FastMTT alone carried the whole deficit (Delphes retained 12% against CMS’s 99%); now 1.01 overall. Unit tests inject a loss at one known stage and assert it is attributed there and nowhere else.
2. **The FastMTT failure, and why the mass must be sampled.** The hadronic prior is zero unless $x_{\min} = (m_{\text{vis}}/m_\tau)^2 \leq 1$; a Delphes τ -jet carries the AK4 jet mass ($\mathcal{O}(5\text{--}20)$ GeV against $m_\tau = 1.777$), so no solution exists and $m_{\tau\tau}$ returns NaN — silently removing $\sim 89\%$ of pairs. Assigning the anchor visible mass fixes it, but it must be *sampled*, not averaged: the mass is a one-sided floor, so a single median relaxes a real constraint for every leg above it, inflating the width $\sim 60\%$ and the mean $\sim 9\%$ while leaving the *mode* unchanged — a mode-gated $Z \rightarrow \tau\tau$ candle would not catch it.
3. **The reference error, twice.** A Delphes GenJet is also a jet and also contains the underlying event, so it is *not* the analogue of GenVisTau. Against a genuine visible τ ($p_{\text{vis}} = p_\tau - p_{\nu_\tau}$, exact since a hadronic decay has one neutrino), the Delphes τ -jet is 17% harder at 25 GeV, falling to $\sim 2\%$ at 250 GeV. The same error appeared independently in the energy-scale map and in the cutflow’s stage-1 acceptance.
4. **Order of application.** Maps measured against CMS-scale quantities must be *applied* at a CMS-scale p_T . Drawing tags on raw jets evaluated a steeply-rising efficiency $\sim 17\%$ too far right; applying the energy scale first moved the analysis-level excess $1.25 \rightarrow 1.18$.
5. **Gen- τ construction check** (`gen_tau_check.py`). Our visible- τ construction scored against CMS’s own GenVisTau on the same events: efficiency 0.991, purity 0.998, reproducing CMS’s stage-1 rate to $\sim 1.6\%$.
6. **m_{vis} and the gen-matched/fake split** (`nsbi_overlay.py --diagnostics --split-gen-matched`). m_{vis} uses neither the p_T^{miss} nor the fit, so it attributes an $m_{\tau\tau}$ offset to one of two disjoint causes; three unit tests pin the properties the reading depends on ($m_{\text{vis}} \leq m_{\tau\tau}$ always; invariant under a $40 \rightarrow 200$ GeV p_T^{miss} change; exactly linear in the τ energy scale). The split normalises both components to the side’s *total*, so the area under the fake curve is the fake fraction. Results in Section 2.1.
7. **Map accuracy audit.** No map is statistics-limited at the configured 2×10^5 anchor events: b -tag 0.25–0.42% per bin, τ_h efficiency 0.43–1.26%, energy scale 0.2–0.3%, p_T^{miss} resolution 0.16–0.21%. What limits a map is the numerator np , not n . Two structural defects do not improve with statistics — the mass quantile grid interpolates across the physically empty region between the $\pi^\pm$ atom and the ρ shoulder, over-populating it $\sim 6\times$; and the 300 GeV flat extrapolation — but the first was measured inert and the second is untested.
8. **Generator copies.** Three of our own errors had one cause: Delphes writes the full Pythia history, so each physical τ appears several times, while CMS GenPart is pruned. The rule: anything that **counts** generator objects must collapse copies first; anything that **matches** them geometrically need not, since copies are collinear.

Reproducing.

```
# NB --max-events is required: derive_maps caps the ANCHOR from the config but reads
# the Delphes side uncapped, so without it this reads the whole production into memory.
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \
    --output cards/tuning/maps_v1.json --max-events 200000
```

```
python -m delphes_pipeline.validation.run_validation --config config.v1.yml
python scripts/tauh_cutflow.py --config config.v1.yml [...] --gen-dr plots/gen_dr
python scripts/nsbi_overlay.py --config config.v1.yml [...] \
    --tautau-only --diagnostics --split-gen-matched
```